INDEPENDENT CHECK · NOT PEER REVIEWED

Do gravitational waves decay into light?

An independent check of D. Blas and J. A. Oller, “Gravitational waves decay in vacuum,” arXiv:2607.24930 (version 2, 31 July 2026; an unpublished preprint)

Prepared for Ethan Van Winkle · 27 October 2026 · From the Science repository, github.com/EthanDJohnson/Science, folder `checks/2026-10-11-blas-oller-gw-decay/`

HOW THIS WAS MADE

The check was run by AI agents (Anthropic’s Claude models), directed from the Science repository. Several independent analyses attacked the paper from different directions, adversarial referees challenged each one, and a final critic re-ran the scripts. Every number below was printed by a script whose output log is kept beside it. Two more reviewer agents then checked this summary against the report and logs, and their corrections are included. Separately, a literature search looked for anyone who had made these points before (Section 10). No physicist has reviewed any of it yet. Read it as a careful draft of a referee report, not as a published result. Where it is uncertain, it says so.

The short version

- **The claim.** Treat a gravitational wave as a coherent state of N gravitons. Two gravitons can fuse into two photons ($g\,g \to \gamma\gamma$). Each fusion is suppressed by G^2, but the rate grows as N^2, so the authors find that gravitational waves decay into photon pairs in empty space. They apply it to binary mergers, the stochastic gravitational-wave background and dark matter, with an extra boost from the photons of the cosmic microwave background (CMB).
- **Their algebra checks out inside their model.** We reproduced their amplitudes by three independent routes, their key coefficient $3/(25\pi)$ exactly, their stochastic-background rate $8\pi G^2\rho^2/15$, and their $\hbar$ bookkeeping.
- **But their model is not a wave travelling away from its source,** for two separate reasons:
  1. The amplitudes are complex numbers whose phases they drop. With the phases kept, the rate is 0.25–0.60 of theirs for the polarisation patterns we tried (0.34–0.60 for realistic quadrupole patterns), and exactly zero for a circularly polarised wave.
  2. The quantum state they use carries an *incoming* spherical wave that collapses onto the source and then re-expands. Most of their probability is generated at that collapse point: about 85% within one wavelength for a spherical shell, and about half within one wavelength (82% within two) for a quadrupole shell.
- **A real outgoing wave makes photon pairs only near its source.** The number made beyond a radius r falls as $1/r^3$. So a wave front does not decay in flight: no “reaction time,” and no depletion along its path. (The tiny depletion of a uniform background, eqs. 14–15, is a different calculation, and it checks out.)

- **Not settled:** production right at the source. Inside about λ/2π (318 km for a black-hole merger, about the binary's own size) nothing was computed. Counting from λ/2π outward, a crude quadrupole estimate already gives 0.3–0.7 of their merger number. So we do *not* claim their merger numbers are orders of magnitude too large. We claim they describe production at the source while it radiates, not a gravitational wave decaying in vacuum.
- **Has anyone said this before?** Not as far as a broad search can tell (Section 10). The paper is an unpublished preprint, still at version 2, with no citations, comments or replies. The closest earlier work is the classic result that a plane gravitational wave far from its source creates no particles (Gibbons 1975; Liu 2007), which B&O themselves accept. Our r^{-3} result extends that to a real, spherical wave. None of our other points turned up anywhere.

1. What Blas and Oller claim

Blas and Oller ("B&O" below) model a gravitational-wave front as a coherent state of N gravitons of frequency ω and bandwidth σ. Their abstract opens: "We show that gravitational waves (GW), treated as coherent graviton states, decay into photon pairs in vacuum. The process, even if suppressed by G^2, is lifted by two effects combined: the expected factor of the graviton number squared, N^2, and the coherence of the wave."

The ingredients are the tree-level amplitudes for two gravitons going to two photons (their eq. 6), with magnitudes $8\pi G\, t^2/s$ and $8\pi G\, u^2/s$ in terms of the Mandelstam variables, summed coherently over the wave's graviton modes (eq. 7). The result is a pair-production rate (eq. 8), in natural units ($\hbar = c = 1$):

$$dP/dt = 3\, G^2\, N^2\, \omega^3\, \sigma^2 / (25\pi) \qquad \text{B\&O eq. 8}$$

From it they derive a "characteristic reaction time" τ (by their eq. 9, the time to the first photon) and a depletion time for the wave (eqs. 9–12); a photon luminosity for compact binaries (their Fig. 2); a rate per volume for a stochastic background, $8\pi G^2 \rho_{GW}^2/15$ (eq. 14); an enhancement by CMB photons already present, a factor $1 + 2f$ with f the CMB occupation number; a cosmological integration over the history of all mergers (eq. 19 and supplement eq. S12); and the analogous decay into ultralight dark-matter scalars (eqs. 16–18).

2. How we checked it

Four analyses, each with its own scripts:

1. **Amplitudes and helicity phases.** We rebuilt the full complex amplitude three ways that share no code: a new amplitude module, a spinor-helicity "double copy" (KLT) closed form, and a covariant route with no spinors. The first two agree on the full complex amplitude to 10^{-12} for random kinematics. The third, whose overall constant is fixed at one point, then reproduces the magnitudes to 10^{-14} head-on and 9×10^{-9} over 200 random configurations, and the convention-free phases exactly.
2. **The space-time structure of the state.** What field the coherent state actually carries, and where in space and time the pair probability is generated.
3. **The $2\pi\delta(0) \to T$ step**, which turns a probability into a rate.

4. **The stochastic background, CMB stimulation and ultralight dark matter.**

Adversarial referees then tried to break each analysis, and a critic re-ran the scripts and checked the report against the logs. We quoted B&O only from their own text, extracted from the PDF.

3. What holds up

Inside B&O's idealised model, the following are right:

- the **magnitudes** of the eq. 6 amplitudes, and the pattern of which ones vanish, provided the labels are read as spin projections on a common axis rather than physical helicities (which matters for polarised waves);
- the **eq. 8 coefficient** 3/(25π), exactly, from their own integral;
- the **stochastic-background rate** (eq. 14), $8\pi G^2\rho^2/15$ per unit volume, for any isotropic unpolarised spectrum, and their conclusion that it is utterly negligible (10^{-110} to 5×10^{-109} photons per cubic metre per second, depending on the background's size);
- the **drop-out of coherent intermediate gravitons**, and the scalar amplitudes and coefficients (7/(450π), 1/16) for the dark-matter channel;
- the **ħ counting**: the pair number contains no ħ at fixed classical wave energy and frequency. It is a quartic functional of the wave's classical strain, so the celebrated N^2 is ordinary classical phase coherence, not a quantum bonus. Its large size comes from the small focal region discussed in Section 5, not from an extra power of N (the incoherent background rate, eq. 14, is also quadratic in graviton number).

The problems are not arithmetic. They are about whether the quantity computed is the one that describes a real gravitational wave.

4. Problem 1: the helicity phases are dropped

Scattering amplitudes are complex numbers, and for g g → γγ their phases depend on the directions of the outgoing photons. B&O's eq. 6 lists magnitudes whose labels are spin projections on a common axis rather than physical helicities. That is fine for the magnitudes, but B&O's eq. 7 then adds the amplitudes as *real positive numbers*. Combinations of them that don't depend on phase conventions have phases of +63.9°, −17.1° and −125.8° (one photon direction, three polarisation tilts) rather than 0°, so no choice of convention makes them all real at once.

Keeping the phases matters. Two angular-momentum channels ($J_z = +4$ and -4) cannot interfere once you integrate over the photons' azimuth, because their cross term goes as cos 8φ and averages to zero. B&O's rule keeps that cross term as a constant. As a result:

- for B&O's own head-on pair (aligned "plus" polarisations) the rate is 6.4 (in the check's units), not B&O's 7.47;
- the rate depends wrongly on the relative orientation of two linear polarisations (5.33 instead of the physical 6.4 for plus × cross);
- read literally, eq. 7 gives a nonzero rate for two right-circular gravitons, where the physical rate is exactly zero.

B&O's "unpolarised isotropic" wave has no unique physical realisation (a smooth polarisation pattern cannot be combed flat over a sphere, the hairy-ball theorem). So we evaluated several real patterns:

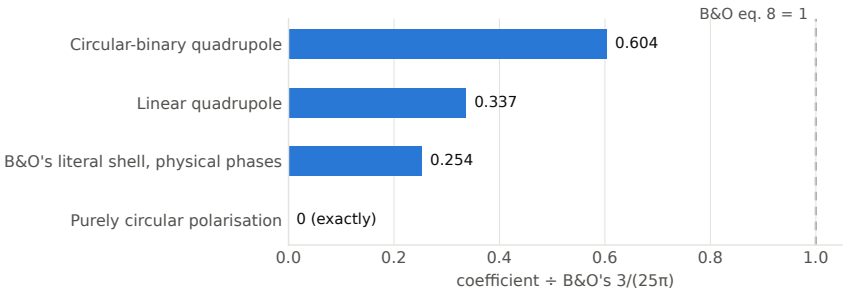


Figure 1. The eq. 8 coefficient for physical coherent shells, as a fraction of B&O's 3/(25π). An independent integrator agrees to 0.01% (to 0.2% for the literal shell, which converges slowly). For the same linear-quadrupole pattern, B&O's own rule gives 0.891 instead of 0.337. Sources: `notes/taskA3b_smooth.log`, `notes/taskA3b_bo_plus_grid2.log`.

Eq. 8 therefore overestimates by 1.7× to 4×, depending on the pattern. A separate point: eq. 5 gives the wave a Gaussian profile, but eqs. 7 and S6 switch to a top hat, which B&O say they do "for convenience." At equal N and equal σ parameter the Gaussian gives a 4π times larger peak rate and a 3.76 times larger total (about 1.5 times at equal rms bandwidth), so the 3/(25π) belongs to the top hat specifically.

5. Problem 2: the wave in the calculation comes in before it goes out

B&O compute an S-matrix probability with the initial state $|f\rangle = S_{\text{source}}|0\rangle$: the coherent state that the source would create, used as the incoming state of free fields. A free field that agrees with the source's outgoing radiation at late times must, at early times, have been an *incoming* spherical wave. That is the Pauli–Jordan (retarded minus advanced) solution. In the calculation the wave converges onto the source's position, focuses there at $t = 0$, and re-expands. Numerically, the field is 100% incoming at early times ($t\omega = -30$) and 50% incoming at the focus. A real source's retarded field is never incoming.

That focus is where B&O's probability comes from:

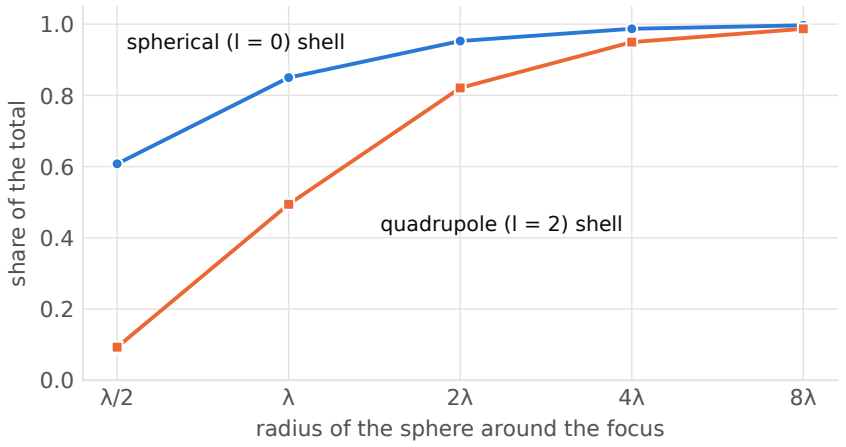


Figure 2. Share of the total pair probability generated inside a sphere of the given radius around the focus, for the free (Pauli–Jordan) field of a Gaussian shell, computed with a scalar stand-in for the graviton field ($\sigma = 0.1\omega$): B&O's spherical shell and, for comparison, a quadrupole shell. For the spherical shell, 85% comes from within one wavelength, and removing everything inside one wavelength leaves 2%. The shares don't add to 100% because the probability is quartic in the field, so inner and outer regions interfere. The quadrupole shell is more spread out but still local (82% within two wavelengths). B&O's "effective volume" $5\lambda^3/(9\pi\varepsilon^2)$, 17.7 λ^3 at $\varepsilon = 0.1$ (7.9 λ^3 at the $\varepsilon = 0.15$ of their examples), is this focal region. Source: `notes/taskB_where_grid.log`.

Why the focus matters: nearly parallel gravitons barely fuse. Two massless particles moving in exactly the same direction have zero invariant mass, $s = 0$. There is no threshold for $g\,g \to \gamma\gamma$, since the photons are massless too, but the amplitude itself shrinks with s: with $s + t + u = 0$, eq. 6 gives $|M| = 8\pi G\, t^2/s \le 8\pi G\, s$, so the cross-section goes as $G^2 s$. Far from its source a wave is locally almost a plane wave, its gravitons nearly parallel, and s is only of order $4\omega/r$ (natural units). Production is strongly suppressed but not zero, which is why the yield falls as a power of r (Section 7) instead of switching off. At the focus, gravitons arrive from every direction, s is large, and fusion is easy. For an exact plane wave, $s = 0$ and nothing is produced.

To be fair to B&O: they note that far from the source the field tends to the outgoing packet, which is true at late times. They also use the Feynman propagator correctly for the internal graviton. The problem is the external state: the S-matrix integrates over all times, including the fictitious incoming half of the wave.

6. A consequence: eq. 8 is a peak rate, not a sustained one

B&O turn the energy-conserving delta function into a time by the textbook replacement $2\pi\delta(0) \to T$. Kept exactly instead, it gives a finite total for the whole packet,

$$P_{\text{total}} = (4/25)\, G^2\, N^2\, \omega^3\, \sigma \text{ pairs},$$

produced with a time profile $R(t) = R(0)\,\text{sinc}^4(\sigma t/2)$, where $R(0)$ is exactly B&O's eq. 8. So eq. 8 is the *peak* rate, sustained for an effective time $4\pi/(3\sigma)$.

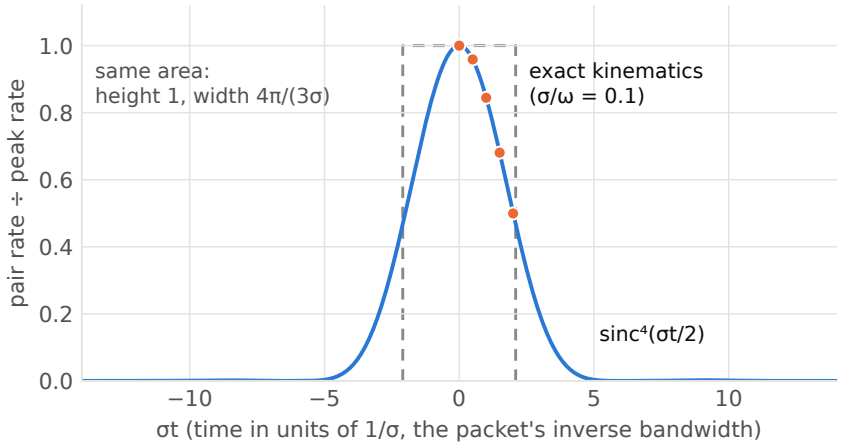


Figure 3. Pair production in B&O's own state happens once, in a burst lasting about four inverse bandwidths ($4\pi/(3\sigma)$; full width at half maximum $4.0/\sigma$). The curve is $\text{sinc}^4(\sigma t/2)$; the dots are an exact-kinematics calculation ($\sigma/\omega = 0.1$), normalised to their own peak, which is 0.96 of eq. 8 at that bandwidth; the dashed rectangle has the same area as the curve. Source: `notes/taskC1.log`.

As an average over the packet, B&O's idealisation is legitimate. What breaks is using it as a constant rate over times much longer than $1/\sigma$. Eq. 9's photon count exceeds the total photon number, $2P_{\text{total}}$, for $t > 4\pi/(3\sigma)$, and in B&O's own model the fraction of a packet that is ever converted is $(4\pi/3)/(\sigma\tau_h)$. For their own example ($E = M_\odot c^2$, $\varepsilon = 0.15$, $\omega = 10^4$ eV) that fraction is 2.4×10^{-36}. That is not depletion.

7. What a real outgoing wave does

To model a physical wave, we kept only the outgoing half of B&O's packet, switched on beyond a radius r_{min}, and estimated the pairs it produces with the in-in (expectation-value) formalism, using a scalar stand-in for the gravitational wave rescaled by a gravitational coherence factor of 0.41, which only one module computes:

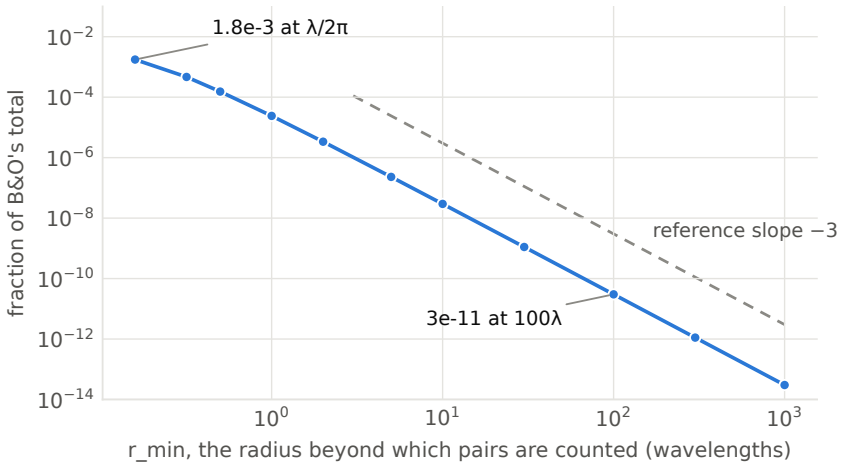


Figure 4. Pairs produced beyond r_{min} by the outgoing wave, as a fraction of B&O's total. The coherence factor of 0.41 shifts the curve but not its slope. The local slope tends to −3.000; smooth switch-on shapes change the prefactor by up to 2.2×, and a window-free attribution is several times higher, with the gap growing only as $\log^2 r$. Sources: `notes/taskB_outgoing.log`, section (h) for the curve, sections (c) and (g) for the slope and the switch-on spread.

The rate while the shell passes radius r falls as r^{-4}, and a quadrupole wave with its near-field terms behaves the same way. This is a formation-length effect, and it undoes the claims that rely on a sustained decay:

- **No reaction time and no depletion of a wave front along its path** (eqs. 9–12; their footnote 4, which pictures photons produced on a "sphere of radius τ").
- **The cosmological integration (eq. 19/S12) overcounts.** It lets each merger radiate photons for the whole propagation time. In B&O's own model the energy per event is the rate times $4\pi/(3\sigma)$, so S12 overcounts by $3\sigma/(4\pi H)$: an order-of-magnitude estimate of about 10^{21} at $\omega = 10^{-10}$ eV today (1.7×10^{21} for a fractional bandwidth $\varepsilon = 0.1$; the factor scales with ε).
- **The "lower bound" from binning is not guaranteed.** B&O split a chirping signal into frequency bins and apply eq. 8 to each, calling the result a lower bound (while noting that neglected terms "increase the decoherence as ϵ grows"). For a bin lasting many wave cycles the exact chirped result is smaller, by about $(\sigma t_\varepsilon)^2/(4\pi^2)$ where t_ε is the time the chirp spends in the bin, so eq. 8 per bin overstates it, by up to about 7,000 inside B&O's own allowed region. The typical allowed bin is short ($\sigma t_\varepsilon \approx 2.7$), where that formula doesn't apply and the comparison was not computed. So "lower bound" is not guaranteed, and how far off the binning is in practice is contested.

Worked example: a GW150914-like merger

$E = 2\,M_\odot c^2$ radiated at 150 Hz over 5 ms (SI units):

Quantity	Value
B&O eq. 8 rate, applied to the merger	1.40×10^{-4} pairs/s

B&O reaction time τ (eq. 10)	3.58×10^{3} s (720,000 times the merger's duration)
B&O rate × 5 ms	7.0×10^{-7} pairs (about 5×10^{2} with the CMB enhancement)
Outgoing wave, pairs made beyond λ/2π (318 km)	1.9×10^{-9} pairs (1.4 with the CMB enhancement)
Outgoing wave, beyond one wavelength (2,000 km)	2.6×10^{-11} pairs (0.02 with the CMB enhancement)
Outgoing wave, beyond ten wavelengths	3.2×10^{-14} pairs
Quadrupole wave with its near-field $1/r^2$ and $1/r^3$ terms, beyond λ/2π	$2.1–4.9\times10^{-7}$ pairs, i.e. 0.30–0.70 of B&O

The last row is why we don't claim B&O's merger numbers are hugely too large: close to the source the numbers are comparable (compare like with like: both columns without the CMB enhancement, or both with it). But $\lambda/2\pi = 318$ km is about the binary's own separation (339 km), inside the source, which none of these models describe. B&O's validity condition $\tau \ll 1/\sigma$ also fails in vacuum and holds only with the CMB enhancement, where $\tau = 4.7$ μs and their "sphere of radius τ" has a radius of 1.4 km, again deep inside the source.

8. Verdict on each claim

"Right arithmetic, wrong quantity" means the calculation is correct but does not describe a gravitational wave travelling away from a real source.

B&O claim	Verdict	Key number
Eq. 6 amplitude magnitudes and zero pattern	**Confirmed**	as spin projections on a common axis; agree to 10^{-10}
Eq. 6 amplitudes usable as real positive numbers (as eq. 7 needs)	**Refuted**	invariant phases +63.9°, −17.1°, −125.8°
Eq. 8 coefficient 3/(25π), within the model	**Confirmed**	exact (sympy); numerical ratio 1.000000000000
Eq. 8 for physical coherent waves	**Right arithmetic, wrong quantity**	0.25–0.60 × eq. 8 for the patterns tried; 0 if circular
Coefficient set by N, ω, σ alone	**Refuted**	depends on profile (4π) and polarisation
$2\pi\delta(0) \to T$	**Confirmed as an average**	eq. 8 is the peak; $T_{eff} = 4\pi/(3\sigma)$
Field "tends asymptotically to the outgoing" packet	**Right arithmetic, wrong quantity**	true late; the in-state is incoming early
Where the eq. 8 probability is generated	**At the focus**	85% within 1λ (spherical shell)
Sustained decay, depletion, "sphere of radius τ"	**Refuted**	yield beyond $r \propto r^{-3}$
Photon luminosity of binaries (Fig. 2)	**Right arithmetic, wrong quantity**	near-source comparison undetermined
Binning gives a lower bound	**Refuted as a guaranteed bound**	overestimates long bins by up to ~7,000; typical bins not computed
Eq. 19/S12: each merger radiates for the whole propagation time	**Refuted**	overcount $3\sigma/(4\pi H)$, $\sim10^{21}$ at 10^{-10} eV
ħ counting (eqs. 11–12) and the N^2 enhancement	**Confirmed, with a caveat**	N^2 is classical phase coherence; its size comes from the idealised focus

Eq. 14, stochastic background $8\pi G^2\rho^2/15$	**Confirmed**	exact for any isotropic, unpolarised spectrum
CMB factor 1 + 2f “in the average sense”	**Confirmed as an approximation**	exact average 5–8% higher
Dark-matter scalar coefficients 7/(450π) and 1/16, within the model	**Confirmed**	physical states 0.20–0.47 of 7/(450π); 0 if circular
Eq. 18, scalar power ~10^{14} W	**Arithmetic right, applicability doubtful**	3.8×10^{13} W; resonant band ~10^{-6} of the merger’s

The full 30-row table, with the script behind every number, is section 3 of `report.md` in the check folder.

9. What is not settled

1. **Production right at the source** (within about λ/2π) is not computed. It needs the binary’s own near-zone field and the coupling of matter to gravitons. This is why we call the comparison with B&O’s binary luminosity undetermined.
2. **The retarded-field argument is analytic only.** That the physical field is the retarded, outgoing one when the source is kept inside the calculation was argued, not checked numerically.
3. **Counting pairs “beyond r_{min}” depends on how the yield is attributed.** Smooth switch-on shapes differ by up to 2.2×, and a window-free attribution is about 6 times higher at one wavelength, with the gap growing slowly (as $\log^2$). The r^{-3} fall-off holds up to those log factors.
4. **There is no unique physical polarisation pattern** for B&O’s isotropic idealisation. The patterns we tried give 0.25–0.60, and a circularly polarised wave gives 0.
5. **The finite-mass scalar channel is incomplete:** its physical interference term was not computed.
6. **The coherence factor of 0.41** used in the outgoing-wave numbers (Figure 4 and the merger table) comes from one module with no independent cross-check. No verdict rests on it.
7. **Eq. 6 and the scalar amplitudes (S10) use different label conventions** (common-axis spin projections and physical helicities). That is harmless for B&O’s unpolarised states but matters for any polarised application.

10. Has anyone else said this?

On 27 October 2026 a team of agents searched for prior work on each of our points. Searchers came at it from five angles: who cites the paper, public discussion, plane-wave physics, coherent-state physics, and the paper’s own versions. An adversarial agent then hunted for anything they had missed, a separate agent opened every source and verified each quote, and a judge weighed it all. The full record is in `checks/2026-10-27-prior-art/blas-oller.md`.

Commentary on B&O: none found. As of 27 October the paper has no citations in INSPIRE, OpenAlex or Semantic Scholar. There is no comment, reply, erratum, journal version or news coverage, and arXiv has no version 3. The only third-party item is an AI-generated review on the Pith site (3 August 2026), which our tools could not read. As paraphrased, it touches the top-hat and binning points only lightly.

Our point	Status	Closest earlier work

Phases dropped (Section 4)	Not found	Textbook background only: helicity states carry convention-dependent phases (Jacob–Wick; Haber's 1994 lectures).
Incoming wave, probability made at the focus (Section 5)	Not found as applied	The identity behind it is standard: retarded − advanced = outgoing − incoming (Kosower, Maybee and O'Connell 2019, eq. 6.26; Cristofoli et al. 2021). B&O even name their kernel "the Pauli-Jordan function," but don't draw the consequence.
Peak rate, finite total (Section 6)	Not found	A QED analogue only: models that ignore pulse duration overestimate light-by-light signals by up to ten times (Dinu et al. 2014).
No decay in flight, r^{-3} (Section 7)	**Partly anticipated**	The plane-wave limit is classic. Gibbons (1975): "a gravitational wave far from its source will propagate without hindrance by quantum effects." Liu (2007, not cited by B&O): such waves "propagate without dissipation via any quantum field theoretic effects." B&O accept this for plane waves. The finite spherical wave, the r^{-3} yield and the cosmological overcount were not found.
Binning "lower bound" (Section 7)	Not found	B&O half-state it themselves: their sentence goes on to say the neglected terms "increase the decoherence as ϵ grows."
Gaussian versus top hat (Section 4)	Not found	B&O disclose the switch ("For convenience, we approximate the previous profile by a narrow-band top-hat distribution"); its effect on the coefficient is ours.

The claim itself is not new either. Jones and Singleton proposed in 2015 that gravitational waves attenuate through $G + G \to \gamma + \gamma$, followed by papers in 2017 and 2019, which B&O do not cite. Those papers predict decay in flight too, so our r^{-3} result cuts against them as well.

An independent recheck. For the prior-art verdict, the judge rebuilt the phase argument (Section 4) from closed-form spinor expressions without our code and reproduced the phases exactly. It also confirmed the Gaussian/top-hat factors (4π and $3\sqrt{(\pi/2)} = 3.76$) analytically. Nothing in the search contradicts the check.

Limits of the search. The arXiv search interface and Semantic Scholar's citation listing were rate-limited throughout, so a very new arXiv paper not yet in INSPIRE could have been missed. Google Scholar, ResearchGate and ADS were unreachable, and private channels such as referee reports and email can't be searched at all. Our own check has been publicly readable on GitHub since 13 October 2026.

Where this could go next

You'll know better than we do. Some options:

- **A sanity check on Problem 2 by someone who works with QFT in curved or radiating backgrounds.** It is the crux for the "decay in flight" conclusion. If the in-state argument is wrong, the focus and r^{-3} findings lose their force, though the phase, profile and finite-total findings stand on their own.
- **Writing to the authors.** The prior-art judge's view is that a polite technical note to Blas and Oller is worthwhile, since the paper is still an unpublished preprint. The phase argument

(Section 4) and the finite total (Section 6) are the quickest for them to check, and the incoming-wave point (Section 5) matters most. A note should credit Gibbons and Liu for the plane-wave limit. A short “Comment on” paper is the other route; posting one on arXiv needs an endorser in the field.

- **Finishing the open piece:** production inside the source region, with a real binary model.
- **In the classroom,** the argument in Section 5 may be the most usable piece. The g g → γγ cross-section grows with the pair’s invariant mass ($\sigma \propto G^2 s$), and nearly parallel gravitons have tiny s, so a wave far from its source makes almost no photon pairs (the yield beyond r falls as r^{-3}).

Reproducing the numbers

Everything is in `checks/2026-10-11-blas-oller-gw-decay/`: `report.md` (the full report, quoting B&O verbatim), `calc/` (about 35 Python scripts) and `notes/` (their output logs). The scripts need numpy, scipy, sympy and mpmath. Fourteen of them import the repository’s `qft_second_order.py`, and every module output a verdict rests on is cross-checked by an independent route (the coherence factor of 0.41, used only in the outgoing-wave numbers, is not). Run them from the check folder, for example `python3 calc/taskC1_exact_Ptot.py`. A few take several minutes. The figures here are drawn by `summary/make_figures.py` from the logged numbers.

Glossary

Coherent state
The quantum state that best resembles a classical wave: a superposition of graviton numbers with a definite average field. A classical gravitational wave is a coherent state with an enormous N (about 3.6×10^{78} for the merger example).

Helicity
A massless particle’s spin along its direction of motion: ±2 for gravitons, ±1 for photons. Circular polarisation means one helicity.

In-out versus in-in
An S-matrix (“in-out”) calculation asks for the probability to go from a state in the far past to one in the far future. An “in-in” calculation asks for the expectation value of an observable, here the number of photon pairs, given the past state. At the order used here the two give the same number for the same initial state. The disagreement with B&O is about which state and field to use (an in-state carrying the Pauli–Jordan field, versus the retarded, outgoing field of a real source), not about in-in versus in-out.

Pauli–Jordan function
The retarded minus the advanced solution. It is a source-free solution of the wave equation, so it can describe a free wave, and it contains an incoming half as well as an outgoing one.

Formation length
The distance a process needs to “complete.” For nearly parallel gravitons the invariant mass is small, the formation length is long, and pair production far from the source is suppressed.

Natural units
ħ = c = 1, so G has units of inverse energy squared and every quantity can be expressed in powers of energy.